

Function-Vector Heads Are Two Populations: Writers and Cancellers in In-Context Learning*

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Abstract

Function-vector (FV) heads (Todd et al., 2024) are typically identified by the *magnitude* of their causal contribution to in-context rule tasks, under the implicit assumption that the top set is a homogeneous functional class. **This assumption fails.** Replacing magnitude-only ranking with a sign-preserving criterion (refined DLA + permutation FDR) and validating each candidate by path patching, we find the FV head population splits into two opposing sub-populations: *writers* push the rule-correct logit up; *cancellers* push it down. The four-condition canonical verdict holds across three model families and six Pythia scales (13/15 cells canonical), and a sign-shuffle rejects homogeneity in 5/6 main cells. The structure is invisible to magnitude-only ranking: on the hierarchical-rule task, Todd’s top-20 captures 64% of cancellers but only 4% of writers; on the modular-rule task, 59% of writers but only 8% of cancellers. Six artefact accounts are ruled out across all 27 canceller (cell, head) pairs across the six main cells: induction overlap, sinks, generic importance, rank-1 copy-suppression, V-cascade, and rank-nearest non-FV controls. Zero-ablating cancellers produces a +0.13 to +0.29 nats logit gain in 6/6 main cells with a directionally consistent +2–7 pp accuracy effect.

1 Introduction

Function-vector (FV) heads (Todd et al., 2024; Hendel et al., 2023; Variengien and Winsor, 2023; Olson et al., 2022)¹ have become a central object of

*Code and data: <https://github.com/henryhyw/function-vectors-two-populations>.

¹Glossary. *FV head*: a function-vector head. *DLA* (direct logit attribution): a head’s signed contribution to the correct–incorrect logit difference (Elhage et al., 2021; Geva et al., 2022). *OV*: the head’s write-direction in the residual stream; *QK*: the head’s attention-source pattern (Elhage et al., 2021). *Path patching* (Wang et al., 2023; Goldowsky-Dill et al., 2023) isolates a head’s *direct* effect by patching its activation while holding downstream paths fixed.

mechanistic interpretability: a sparse set of attention heads whose causal contribution to in-context-learning (ICL; Brown et al., 2020) rule tasks is large enough to support targeted steering, ablation, and circuit-level claims.

The standard pipeline ranks these heads by the *magnitude* of a causal-contribution metric and treats the top set as one homogeneous functional class. Implicit in this recipe is a sign assumption: each high-magnitude head pushes the model *toward* the demonstrated rule. Under this assumption, magnitude is identity — one number per head.

We show, across the Pythia ladder (410M–12B, six sizes) and two ICL rule tasks plus three spot-check models from different families (Qwen2.5-{1.5, 7}B, GPT-2-medium), that this assumption fails empirically. The high-magnitude population splits cleanly into two opposite roles: *writers* (refined-DLA > 0, push the rule-correct token’s logit up) and *cancellers* (refined-DLA < 0, push it down). The split is large, reproducible across cells, and not explained by attention-sink, induction-overlap, copy-suppression, V-cascade, generic-importance, or rank-nearest non-FV head confounds. We lead with the population finding; the L11.H4 case study (§4.7) is one illustrative head.

Approach. For each (model, task) cell we identify the FV head set via refined DLA, validate causal effects with path patching, partition by sign of direct effect into $\mathcal{W}$ and $\mathcal{C}$, and run a four-condition canonical verdict ($W_{\text{shift}} < 0$, $C_{\text{shift}} > 0$, opposite signs, joint attenuated). The attention-sink confound is ruled out via per-source DLA bucket attribution, joined by five further population-level rule-outs (induction-overlap TOST, generic-importance ratio, rank-1 copy-suppression (McDougall et al., 2023), V-cascade (Wang et al., 2023), rank-nearest non-FV head). A complementary single-head case study localises the dominant can-

celler L11.H4 in Pythia-410M with three orthogonal interventions (V-shuffle, OV singular spectrum, and V-composition with the dominant writer) and asks which standard mechanistic template it instantiates.

Related work. The FV-head literature (Todd et al., 2024; Hendel et al., 2023; Variengien and Winsor, 2023; Subramani et al., 2022) identifies a sparse set of heads whose causal contribution to a rule task supports steering and ablation, and ranks them by the *magnitude* of that contribution — a pipeline that implicitly treats the top set as one functional class, the assumption this paper overturns. Sign-opposed heads have appeared before, but only as *single-task, hand-built circuits*: IOI’s negative name-movers (Wang et al., 2023) and GPT-2 L10.H7’s copy-suppression (McDougall et al., 2023), each localised to one task by targeted circuit analysis. We lift the same sign-opposition to a population-level property holding across tasks, scales, and architectures (comparison in §5, Table 5).

Methodologically we use standard tooling — direct-logit attribution (Geva et al., 2022; Elhage et al., 2021), path patching (Wang et al., 2023; Goldowsky-Dill et al., 2023), causal mediation (Vig et al., 2020; Meng et al., 2022), and the OV/QK decomposition (Elhage et al., 2021). Sparse-autoencoder features (Bricken et al., 2023) and circuit-discovery catalogues (Hanna et al., 2023; Nanda et al., 2023; Conmy et al., 2023; Syed et al., 2024; Marks et al., 2025) are complementary; we instead re-examine the *sign* of effects within an existing head inventory.

Contributions. (1) **Population finding.** The FV head set splits into two opposing functional roles: *writers* push the rule-correct logit up; *cancellers* push it down. The split is causal (6/6 main lesions; 13/15 with transfer) and resists six artefact accounts — induction overlap, BOS / format-prefix sinks, generic importance, rank-1 copy-suppression (McDougall et al., 2023), V-cascade (Wang et al., 2023), and a rank-nearest non-FV control — on all 27 canceller (cell, head) pairs. (2) **Single-head mechanism candidate.** L11.H4 (Pythia-410M, dominant canceller in both rule tasks) is a *distributed-bias content reader*: a single head whose role flips across templates as “suppress what was demonstrated” predicts, but whose mechanism fits neither rank-1 copy-suppression nor V-cascade. We do not extrapolate this template to

Model	arch.	L	H/layer	d_{model}	d_h
Pythia-410M	GPT-NeoX	24	16	1024	64
Pythia-1B	GPT-NeoX	16	8	2048	256
Pythia-1.4B	GPT-NeoX	24	16	2048	128
Pythia-2.8B	GPT-NeoX	32	32	2560	80
Pythia-6.9B	GPT-NeoX	32	32	4096	128
Pythia-12B	GPT-NeoX	36	40	5120	128
Qwen2.5-1.5B	Llama-style	28	12	1536	128
Qwen2.5-7B	Llama-style	28	28	3584	128
GPT-2-medium	Conv1D	24	16	1024	64

Table 1: Models studied. The first three rows form the main mechanism pipeline (Pythia 410M / 1B / 1.4B); the remaining six are extension cells used for behavioural-verdict transfer (§4.6).

the full canceller population. (3) **Calibration-free intervention.** Zero-ablating cancellers raises the correct-incorrect log-prob difference by +0.13 to +0.29 nats with CI excluding 0 in 6/6 main cells, and shifts accuracy by +2 to +7 pp (CI-confirmed on hier-410M). This is *not* the Todd-style aggregation route, which we show is the wrong operation for the purpose (§5).

2 Setup

Models. The main pipeline studies three Pythia checkpoints (Biderman et al., 2023) (410M, 1B, 1.4B), loaded in `fp32` with non-fused (eager) attention so per-head weights and post-softmax outputs remain exposed to forward hooks; see App. A.2 for the exact HuggingFace flag. Pythia provides a single architecture family across an order-of-magnitude scale range, isolating the scale axis from architectural confounds. §4.6 extends to the rest of the Pythia ladder (Pythia-2.8B/6.9B/12B; bf16 on the $\geq 6.9\text{B}$ sizes), and adds three single-task spot checks in different model families: Qwen2.5-1.5B/7B (Llama-style self-attention) and GPT-2-medium (Conv1D-style attention) on modular only, covering non-NeoX attention architectures.

Tasks. Two ICL rule-following tasks share a 4-shot surface form. Each prompt is a sequence of $K = 4$ demonstrations followed by a query, written as $f_0, f_1, Y \dots f_{0q}, f_{1q}$, where the model must emit the label for the query pair. Feature pairs $(f_0, f_1) \in \{0, \dots, 7\}^2$ are sampled i.i.d. across demonstrations within a prompt; labels are $Y \in \{A, B\}$; a hidden rule $z \in \{0, 1, 2, 3\}$ is fixed within a prompt. A concrete *hierarchical* prompt with $z = 2$ (rule $f_0 + f_1 > 7 \Rightarrow \text{label A}$) looks like:

3, 5, A 1, 2, B 4, 5, A 2, 3, B 6, 3, $\rightarrow$ A

The two rule families differ in mechanism: *Hierarchical*: z selects one of $\{f_0 > 3, f_1 > 3, f_0 + f_1 > 7, f_0 = f_1\}$. *Modular*: z selects modulus $m \in \{2, 3, 5, 7\}$ (coprime, so no rule is a subset of another); the label is $1[(f_0 + f_1) \bmod m = 0]$. They share surface form but differ in rule type, so a head population that survives both is recruited by the rule, not the template.

Notation. Let (L, H) index a head, with $h_{(L,H)}(x) \in \mathbb{R}^{d_h}$ its output at the final query-token position before $W_O^{(L)}$. W_U is the unembedding matrix; γ is the final-LayerNorm gain; $r(x)$ is the pre-LN residual. We aggregate correct/incorrect-label probabilities over leading-space variants (e.g. $\{ \text{'A'}, \text{'A'} \}$ for label A) and define the unembed contrast $u(x) = W_U[y_+] - W_U[y_-]$, the readout $\Delta\ell(x) = \log p(y_+ | x) - \log p(y_- | x)$, and the per-pair swing $\Delta\ell(x_c) - \Delta\ell(x_r)$ between correct- and rule-flipped prompts.

Three prompt sets are sampled per (model, task) cell, with seeds fixed in advance: 192 balanced *discovery* prompts (used for refined-DLA, seed 42), 200 paired *path-patching* prompts (seed 43), and 500 paired *evaluation* prompts (seed 44) used for group lesions, cross-task transfer, and specificity. Logits are aggregated over the leading-space label variants above. Full prompt examples and the format-prefix string are in App. A.2.

3 Methodology

The pipeline runs independently per (model, task) cell: *discovery* screens candidate FV heads (§3.1–3.2), group lesions *arbitrate* the cancellation signature against two nulls (§3.3), and a *mechanism* and *robustness* battery (§3.4–3.5) tests that the writer/canceller labels are intrinsic rather than artefacts of the discovery data. All thresholds and the 2×3 Pythia grid are fixed in advance; statistical conventions (paired-prompt percentile bootstraps with $B = 10,000$, BH-FDR (Benjamini and Hochberg, 1995) at $q = 0.10$) and a full per-test inventory are in App. A.1.

3.1 Refined Direct Logit Attribution

We attribute $\Delta\ell$ to head (L, H) by projecting its post- W_O contribution onto the unembed difference under a frozen-LN linearisation (Geva et al., 2022; Wang et al., 2023):

$$\widehat{\text{DLA}}_{(L,H)}(x) = u(x)^\top \left(\gamma \odot \frac{c_{(L,H)}(x) - \overline{c_{(L,H)}}(x)}{\sigma(r(x)) + \varepsilon} \right), \quad (1)$$

where $c_{(L,H)}$ is the head’s post- W_O contribution to the residual stream, $u(x) = W_U[y_+] - W_U[y_-]$ is the correct-minus-incorrect unembed direction, $\overline{c_{(L,H)}}$ is the discovery-batch mean (the LN linearisation reference), and $\sigma(r(x))$, γ are the standard LN scale and gain.² Per-head signed means are computed on the discovery batch (seed 42, $N = 192$); a 20-seed label-permutation null with BH-FDR at $q = 0.10$ ($L \geq 1$) defines the FDR-passing set. The final FV-candidate set is its union with the top- K heads by mean $|\widehat{\text{DLA}}|$, ensuring a tractable short-list when the FDR-passing set is sparse (exact K rule in App. A.2).

3.2 Path Patching with Ratio-of-Means

We form 200 paired prompts (x_c, x_r) sharing query (f_{0q}, f_{1q}) but differing in rule (Wang et al., 2023; Goldowsky-Dill et al., 2023). After filtering pairs with $|\Delta\ell^{\text{full}}| < 0.5$, the ratio-of-means

$$\text{direct}_{(L,H)} = \frac{\mathbb{E}[\Delta\ell_i^{\text{direct}}]}{\mathbb{E}[|\Delta\ell_i^{\text{full}}|]} \quad (2)$$

(and analogously *total*, *indirect*) is bootstrapped ($B = 10,000$) for 95% CIs and a two-sided p_{direct} . PP p -values are BH-corrected at $q = 0.10$. The $\pm 5\%$ gate is fixed in advance; sensitivity to gate width ($\pm 3\%$, $\pm 7\%$, $\pm 10\%$) is reported in App. A.3. We also impose the two-hop ordering $L_W < L_C$ for any (W, C) pair entered into edge-level analysis.

Operational definition of writers and cancellers.

$\mathcal{F}$ is the sign-preserving FV candidate set from refined-DLA screening (§3.1): unlike Todd et al. (2024)’s top- K by signed AIE,³ our criterion admits heads with both signs of direct effect. A head $(L, H) \in \mathcal{F}$ is a *writer* if its direct PP effect is $\geq +5\%$ of the swing, a *canceller* if $\leq -5\%$, and *weak* otherwise; only writers $\mathcal{W}$ and cancellers $\mathcal{C}$ are entered into the group lesion (weak heads are listed in $|\mathcal{F}|$ for context but excluded from ablations). Per-cell counts appear in Table 3.

Asymmetric overlap with Todd’s magnitude-only set. Todd et al. (2024)’s mean-ablation top-20 captures only one of the two populations per task family (Table 2; full 6-cell breakdown in App. C.2). Magnitude-only ranking systematically surfaces

² $\varepsilon = 10^{-5}$ is the LN numerical stabiliser. The additive identity $\sum_{L,H} \widehat{\text{DLA}} + (\text{MLP, embed, biases}) = \Delta\ell$ is audited end-to-end (rel. tol. 5%).

³Cancellers have positive but smaller AIE than writers and are systematically missed by the AIE-ranked top- K ; we use “magnitude-only” as shorthand for this selection rule.

whichever sub-population locally dominates and obscures the other; only sign-aware screening recovers both.

Table 2: **Todd top-20 overlap, pooled across 6 main cells.** Magnitude-only ranking surfaces opposite populations on the two task families.

task	W in Todd-20	C in Todd-20
hierarchical	1/27 (4%)	9/14 (64%)
modular	16/27 (59%)	1/13 (8%)

3.3 Group Cancellation Lesion: Headline Causal Test

For each ablation strategy $s \in \{\text{zero, mean}\}$ on $n_{\text{eval}} = 500$ paired prompts we ablate (a) writers $\mathcal{W}$, (b) cancellers $\mathcal{C}$, and (c) their union, bootstrapping 95% CI per group ($B = 10,000$).

The canonical cancellation-circuit verdict requires four conditions:

- (i) $|\Delta\ell_{\mathcal{W}}| \geq 0.10$ with CI excluding 0,
- (ii) $|\Delta\ell_{\mathcal{C}}| \geq 0.10$ with CI excluding 0,
- (iii) $\Delta\ell_{\mathcal{W}}$ and $\Delta\ell_{\mathcal{C}}$ have opposite signs,
- (iv) $|\Delta\ell_{\text{both}}| < |\Delta\ell_{\mathcal{W}}| + |\Delta\ell_{\mathcal{C}}|$ (joint attenuation).

Sub-canonical patterns — writer-dominant, canceller-dominant, opposite-signs-without-attenuation, and co-writer — are recorded so non-canonical outcomes remain interpretable.

Sign-shuffle null. Conditions (i)–(iii) are mechanically forced by sign-partitioning; only joint attenuation (iv) is diagnostic. We test it with a sign-shuffle null: for $n_{\text{perm}} = 10,000$ draws we randomly partition $\mathcal{W} \cup \mathcal{C}$ into two subsets matching $|\mathcal{W}|, |\mathcal{C}|$ and record the contrast $\Delta\ell^{\text{permW}} - \Delta\ell^{\text{permC}}$ under zero-strategy ablation. The empirical two-sided p is the fraction of permutations with $|\text{contrast}|$ above the observed, floored at $1/n_{\text{perm}} = 10^{-4}$; FDR control uses the empirical p (Gaussian back-stop in Table 8). A complementary |DLA|-matched head-randomised control confirms that the structural attenuation is FV-specific in magnitude (App. E.1).

3.4 QK Source Decomposition

Each head’s last-position attention mass is partitioned into five buckets (BOS, format prefix, demonstration input, demonstration label, query input) and averaged over $n_{\text{prompts}} = 32$ eval prompts to produce a per-head bucket distribution. We report the per-bucket mean attention mass within $\mathcal{W}$ and $\mathcal{C}$ and their per-bucket difference (canceller minus writer) per cell. We ask whether the canceller-

minus-writer sign is consistent across cells on the demonstration-label and format-prefix buckets.

3.5 Auxiliary tests

Six auxiliary tests support §4.4 (full specs in App. A.1, Table 6). *Split-half*: 5 random 50/50 prompt splits; a cell passes if $\geq \lceil n/2 \rceil$ splits reach canonical on the held-out half. *Cross-task transfer*: W/C labels from cell A applied to cell B ’s eval prompts under a relaxed (i)–(iii) variant (sign persistence, not the strict attenuation gate). *Within-layer OV null*: per (W, C) pair, ≥ 100 random same-layer non-FV pairs define a one-tailed cosine null; a cell passes if $\geq 30\%$ of pairs are significantly anti-aligned at $p \leq 0.05$. *TOST equivalence*: margin $\Delta = \sqrt{\text{Var}_{\text{hg}}}$ from the hypergeometric under random allocation, used for the induction-overlap rule-out. *Specificity*: zero-ablate each canceller on rule and random-token sequences; rule-specific iff $\mathbb{E}[\Delta\text{NLL}^{\text{rule}}]/\mathbb{E}[\Delta\text{NLL}^{\text{rand}}] \geq 5$. *Induction-head overlap*: top-10 heads by prefix-matching (Olsson et al., 2022) intersected with $\mathcal{W} \cup \mathcal{C}$, compared against the hypergeometric expectation.

4 Results

The FV head set has size $|\mathcal{F}| \in [12, 23]$ across the 6 main cells. Since conditions (i)–(iii) of the canonical verdict are mechanically forced by sign-partitioning, the load-bearing evidence is the canceller subgroup’s causal effect (mean 5.2× larger than rank-nearest non-FV controls), the sign-shuffle null on joint attenuation, the structural separation of writers and cancellers, and the task-asymmetric overlap with Todd’s top- K — developed in turn below.

4.1 Cancellers suppress the correct-label logit

Ablating the canceller subgroup shifts the readout +0.13 to +0.29 nats *toward* the correct label in 6/6 cells; ablating writers shifts −0.25 to −0.99 nats *away* from it (Table 3; the 6 main cells in Figure 1). The four-condition canonical verdict (§3.3) holds in 11/12 outcomes across 6 cells × 2 ablation strategies (zero, mean); the single miss is (mod-1.4B, mean), where the canceller magnitude (+0.09) sits just below the 0.10 threshold. Table 4 reports the per-cell verdicts under six independent tests, so partial failures (cross-task hier→mod-1B, mean-strategy mod-1.4B, OV gate hier-1B + mod-1.4B) are surfaced rather than buried.

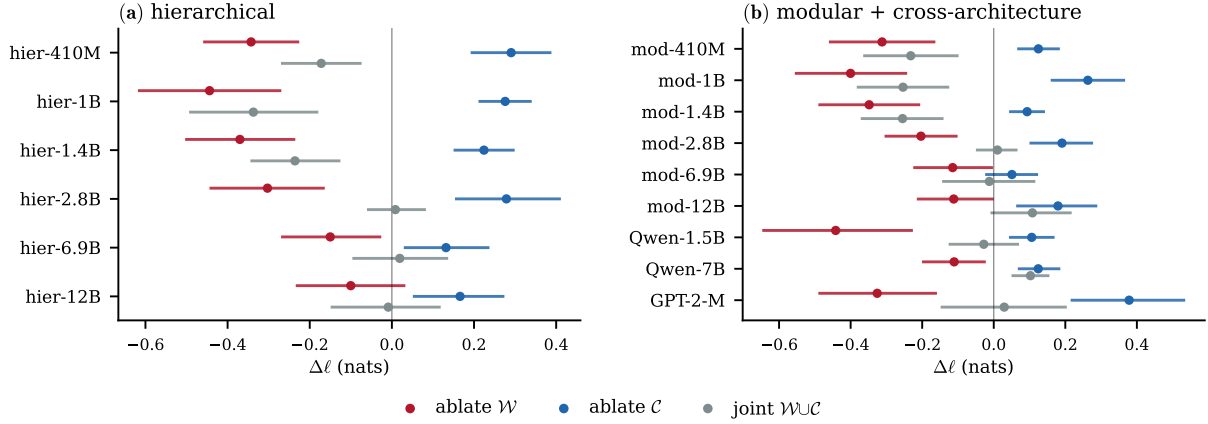


Figure 1: **Function-vector heads are two populations.** Mean-ablation logit shifts (BCa 95% CI) for writers, cancellers, and their joint, per cell. Writer ablation shifts the readout away from the correct label, canceller ablation toward it, and joint ablation attenuates. The four-condition canonical verdict holds in 13/15 cells. (a) Hierarchical (6 Pythia cells, 410M–12B); (b) Modular + cross-architecture (9 cells: Pythia mod 410M–12B, Qwen2.5- $\{1.5, 7\}$ B, GPT-2-medium).

Magnitudes and budgets. Joint ablation attenuates the writer effect by mean 48% (range 21–100%). Writers outnumber cancellers 2:1 with $\sim 1.8\times$ larger per-head magnitudes; per-head canceller shift is uniform (CV 0.19 vs. 0.56). Joint $\mathcal{WUC}$ ablation on mod-410M yields +8.4 pp accuracy (CI [+2.3, +14.5]; the only joint effect with CI excluding 0). Per-cell budgets in App. B.3.

Geometric independence. Writer and canceller mean OV directions are nearly orthogonal in the residual stream (perpendicular fraction mean 0.96, 95% CI [0.91, 0.99]; App. D.4). Cancellers therefore write to a near-orthogonal subspace while still exerting a negative direct effect on u , which lets the two populations co-exist at the same readout and explains why joint ablation is attenuated rather than fully additive.

The verdict survives a sign-shuffle null. Because the W/C sign definition mechanically forces sign-balance, the appropriate null is 10,000 random sign assignments *within* $\mathcal{WUC}$ (zero-strategy ablation). The observed $|\mathcal{W}-\mathcal{C}|$ contrast exceeds the null 95th-percentile in 5/6 cells, with empirical $\text{BH-}q \leq 0.014$ on those five; mod-1.4B is the single boundary case at $p_{\text{emp}} = 0.104$ ($p_{\text{Gauss}} = 0.088$, Table 8). Discussed in Limitations.

A per-cell verdict matrix summarises six independent tests (zero 4c 6/6, sign-shuffle 5/6, split-half 6/6, cross-task 5/6); full breakdown in App. B.2. Per-cell magnitudes are reported in Table 3 (below, following §4.4).

4.2 Writers and cancellers read different prompt positions

QK-source decomposition rules out the hypothesis that cancellers are sign-flipped writer copies. Writers and cancellers read structurally different prompt positions, with the same direction of difference in 6/6 cells on the two load-bearing buckets (Figure 2; per-cell numerics in App. C.3, Table 12). Writers concentrate 34–52% of attention mass on demonstration labels and 28–44% on BOS; cancellers shift 0.09–0.17 of mass off the label position and gain 0.08–0.18 on format-prefix tokens (15–29% of canceller mass vs. 4–17% for writers). Cancellers are not the same heads with a sign flip.

4.3 Mechanism is content, not sinks

The format-prefix pattern raises a sink confound: cancellers might be format sinks whose effect is an artefact of where attention lands rather than what they read. Per-source DLA bucket attribution rules this out. We split each canceller’s direct effect by source bucket and label it *content* when its negative DLA is dominated by demonstration-content buckets, *sink* otherwise. 20 of 27 (74%) cancellers are content-driven, and they carry the headline magnitude (−0.10 to −0.23 nats); the 7 sink heads cluster near zero (−0.02 to −0.09 nats) and reappear cross-task at fixed scale, consistent with format readers rather than rule-specific suppressors. Per-cell numerics in App. C.4, Table 13; an independent V-shuffle protocol on the strongest canceller per cell confirms content-driving in 6/6. L11.H4 is fully content-driven ($\sim 100\%$ demonstration-label sourced in both 410M cells), foreshadowing the

single-head mechanism analysed in §4.7.

4.4 Robustness and specificity

Stability. Split-half ($5 \times 50/50$ prompt splits) is verdict-stable on all 5 seeds in 6/6 cells. Cross-task transfer (W/C labels from a source cell on a destination cell’s eval prompts) is canonical in 5/6 ordered pairs; $\mathcal{W} \cap \mathcal{W}$ Jaccard 0.45–0.70 and $\mathcal{C} \cap \mathcal{C}$ 0.43–0.60 (partial overlap, consistent with the task-conditional populations of §4.5); zero role flips at any scale (App. B.1).

Specificity. 19/27 (70%) cancellers are rule-specific (rule-NLL drop $\geq 5\times$ random-NLL drop; per-cell median ratio 4.8–62 $\times$, up to 151 $\times$ for L11.H4 on mod-410M), trending up with scale (410M: 5/9, 1B: 6/10, 1.4B: 8/8; Cochran–Armitage $p = 0.049$). No above-chance overlap with the induction top-10.

Null comparisons. Within-layer OV anti-alignment holds in 4/6 cells (all 5/5 same-layer pairs significant on mod-410M; App. B.4). Edge-level: real (W, C) pairs are significantly stronger than random layer-respecting pairs in 6/6 cells (Mann–Whitney BH- $q \leq 1.4 \times 10^{-11}$; App. B.4).

Layer geometry. Cancellers cluster in early-to-mid layers and 23/27 have an upstream writer in their own cell, ruling out late-stage suppression (Figure 3). Maximum writer layer reaches $L = 21$ –23 in 24-layer Pythia; canceller layers stop at $L = 19$.

Named-circuit rule-outs. Three named circuits are tested as universal explanations of cancellation and ruled out (full numerics in App. C.7). *Rank-1 copy-suppression* (McDougall et al., 2023): 0/27 cancellers exceed the top-1 OV Frobenius share of 0.5 (observed max 7.5%). *V-cascade* (our shorthand for V-composition in the S-Inhibition pathway of Wang et al., 2023): ruled out in 4/6 cells (2/6 active counterfactual on the dominant upstream writer, 2/6 vacuous with no upstream writer at $L < L_C$); a weak partial residual cascade remains in the other 2/6. *V-shuffle*: permuting the head’s value vectors across source positions collapses cancellation by 86–113% in 6/6 cells, marking the mechanism as content-driven rather than position-driven. With the W vs. C OV near-orthogonality reported above (App. D.4, Table 21), no single named template explains the canceller population.

4.5 Cross-template transfer to vocabulary ICL

We transfer Pythia-410M W/C labels to two vocabulary-ICL templates (antonym, country-capital): canonical in 3/4 pairs (App. C.6). The 1/4 “failure” (hier $\rightarrow$ antonym) flips cancellers to super-writers ($\Delta\ell = -1.56$ nats), the exact pattern copy-suppression (McDougall et al., 2023) predicts when the demonstrated label *is* the answer — the W/C role is task-conditional, not intrinsic.

4.6 Scale and cross-architecture spot checks

The verdict extends to the rest of the Pythia ladder (2.8B/6.9B/12B) and three cross-architecture spot checks (Qwen2.5- $\{1.5, 7\}$ B, GPT-2-medium; modular only, App. B.5, Table 9). All 9 extension cells pass the signed-direction condition; 6.9B mod and 12B hier shift to PARTIAL (one CI grazes 0). Combined with the main cells: 13/15 canonical, 2/15 partial, 0/15 sign-flipped.

4.7 Mechanistic case study of L11.H4

The per-source decomposition of §4.3 identifies L11.H4 as the dominant canceller in both Pythia-410M rule cells, with $\sim 100\%$ of its negative DLA sourced from demonstration-label tokens. We narrow its mechanism with four interventions (numerics in App. C.8).

Content reader (V-shuffle + OV spectrum). Permuting the head’s value vectors across source positions collapses cancellation by $\sim 82\%$ (hier) / $\sim 53\%$ (mod): the head reads source content, not a fixed direction. The OV singular spectrum of $W_O W_V$ is consistent — top-1 Frobenius share 2.8%, top-10 25%, no rank-1 plateau — ruling out rank-1 copy-suppression (McDougall et al., 2023).

V-composition. Ablating the dominant upstream writer L10.H9 leaves L11.H4’s DLA unchanged, ruling out S-inhibition V-cascade (Wang et al., 2023).

Cross-template sign-flip. L11.H4’s solo ablation flips sign across templates (Figure 4): canceller on hier and mod, writer on antonym, null on country-capital. The role is task-conditional — consistent with “suppress what was demonstrated” rather than unconditional logit suppression.

L11.H4 is therefore a *distributed-bias content reader*, distinct from both standard templates. We do not extrapolate this single-head characterisation to the full canceller population.

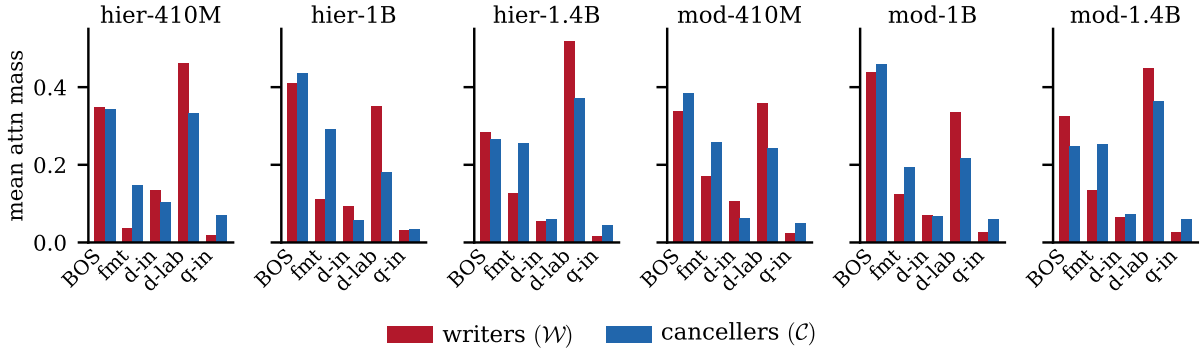


Figure 2: **Cancellers re-route attention from labels to format.** Mean per-head attention mass at the readout token for writers (red) vs cancellers (blue), per bucket, one panel per cell. Writers concentrate on demonstration labels; cancellers shift mass to format-prefix tokens — the direction of difference is consistent in 6/6 cells (per-cell mass in App. C.3, Table 12).

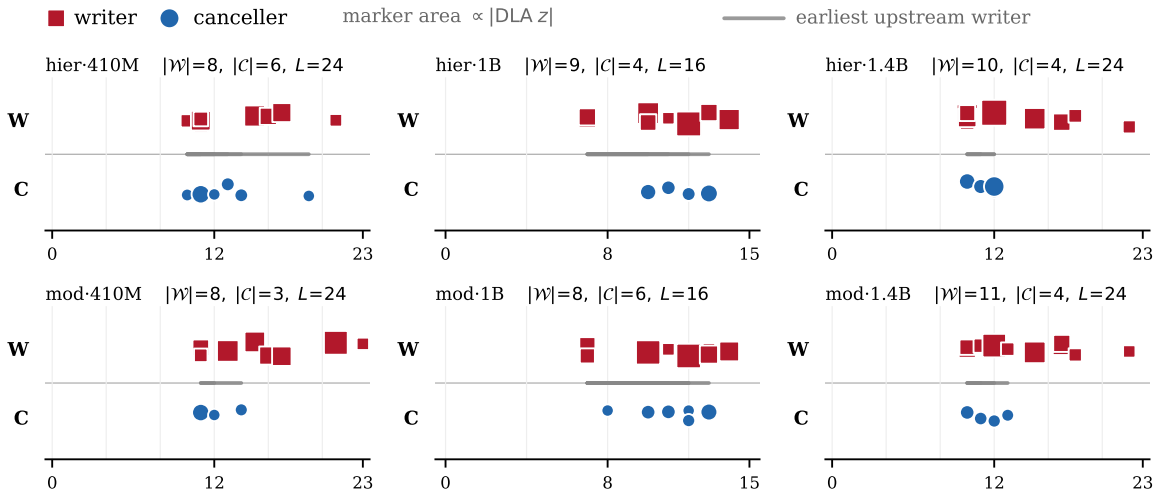


Figure 3: **Layer distribution of writers and cancellers.** Marker area $\propto z$ -scored $|\text{DLA}|$; grey lines connect each canceller to its earliest upstream writer. Cancellers cluster in early-to-mid layers and 23/27 have an upstream writer, ruling out late-stage suppression.

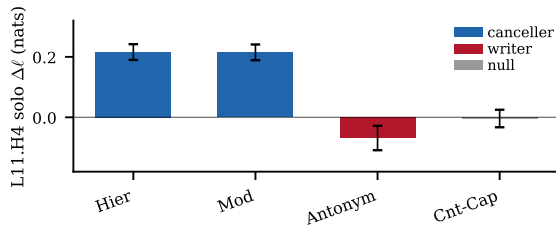


Figure 4: **L11.H4 case study: cross-template sign-flip.** Solo-ablation $\Delta\ell$ on the four templates (paired bootstrap 95% CI). Blue = canceller, red = writer, grey = null. The head’s address transfers across all four templates but its role is task-conditional.

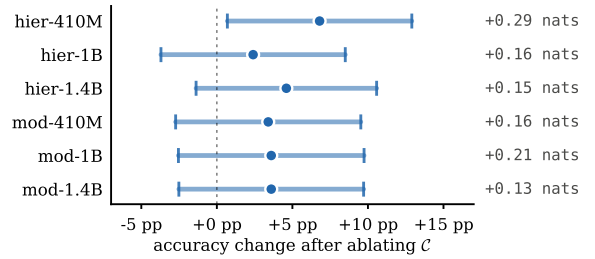


Figure 5: **Canceller ablation: actionable consequence.** Per-cell accuracy delta after zero-ablating $\mathcal{C}$, with paired-bootstrap 95% CI; logit-shift magnitude annotated on the right. Sign is positive in 6/6 main cells; CI excludes 0 on hier-410M. App. D.3 Table 20.

5 Discussion

The FV literature treats high-magnitude heads as one functional class. We split that population into opposing sub-circuits at the same readout, supported by five independent lines: sign-shuffle re-

jects homogeneity (5/6 cells), QK reads differ (6/6), OV directions are near-orthogonal, 23/27 cancellers sit downstream of a writer, and L11.H4 flips canceller→writer on antonym as “suppress what was demonstrated” predicts.

Table 3: **Group lesion (zero strategy).** Writer ablation shifts the readout -0.25 to -0.99 nats away from the correct label; canceller ablation $+0.13$ to $+0.29$ nats toward it; joint attenuates the writer effect by mean 48% (median 41%). Four-condition canonical (§3.3) holds 6/6 zero-strategy, 11/12 mean-strategy (mod-1.4B mean canceller below 0.10; App. B.2). $|\mathcal{F}|$ is the FV candidate set; only $\mathcal{W} \cup \mathcal{C}$ are ablated (weak heads with $|\text{direct\%}| < 5\%$ excluded).

Task	Model	$ \mathcal{F} $	$ \mathcal{W} $	$ \mathcal{C} $	$\Delta\ell_{\mathcal{W}}$	$\Delta\ell_{\mathcal{C}}$	$\Delta\ell_{\text{both}}$	verdict
hier	410M	22	8	6	-0.25	$+0.29$	$+0.00$	canonical
hier	1B	23	9	4	-0.56	$+0.16$	-0.43	canonical
hier	1.4B	19	10	4	-0.32	$+0.15$	-0.13	canonical
mod	410M	12	8	3	-0.39	$+0.16$	-0.14	canonical
mod	1B	16	8	6	-0.99	$+0.21$	-0.78	canonical
mod	1.4B	19	11	4	-0.49	$+0.13$	-0.38	canonical

Table 4: **Per-cell verdict matrix across six independent tests.** $\checkmark$ pass, $\mathbf{x}$ fail, $\circ$ boundary. The cancellation signature holds in 6/6 cells under the strongest test (zero-strategy 4-condition); secondary tests rule out role-flipping, induction overlap, and writer/canceller equivalence on 4–6 of 6 cells. FWER analysis in App. B.2.

Cell	Zero 4c	Mean 4c	Sign-shuf q	Split-half	Cross-task 4c	OV gate
hier-410M	$\checkmark$	$\checkmark$	$\checkmark$	$\checkmark$	$\checkmark$	$\checkmark$
hier-1B	$\checkmark$	$\checkmark$	$\checkmark$	$\checkmark$	$\circ$ W-dom	$\mathbf{x}$
hier-1.4B	$\checkmark$	$\checkmark$	$\checkmark$	$\checkmark$	$\checkmark$	$\checkmark$
mod-410M	$\checkmark$	$\checkmark$	$\checkmark$	$\checkmark$	$\checkmark$	$\checkmark$
mod-1B	$\checkmark$	$\checkmark$	$\checkmark$	$\checkmark$	$\checkmark$	$\checkmark$
mod-1.4B	$\checkmark$	$\circ$ W-dom	$\circ$ ($q=0.104$)	$\checkmark$	$\checkmark$	$\mathbf{x}$
pass	6/6	5/6	5/6	6/6	5/6	4/6

Table 5: **Sign-opposed attention in prior work vs. this paper.** We lift the phenomenon from single-task hand-built circuits to a structural FV-population property.

	Scope	Mechanism	Detection
IOI neg. name-mover (Wang et al., 2023)	single task (IOI)	hand-built circuit	task-specific analysis
GPT-2 L10.H7 (McDougall et al., 2023)	single head, copy task	rank-1 copy-suppression	task-specific analysis
W/C popu-lation (this paper)	population, 13/15 cells, 3 architectures	distributed; no single template fits	sign-aware DLA + path patching

Connection to FV identification. The lift from single-task circuits to a population property is detailed in Table 5 (App. C.5). The practical consequence is that Todd’s top- K is task-conditionally biased toward the locally-dominant sub-population (hier C 64% vs. W 4%; mod W 59% vs. C 8%), so any aggregation built on this ranking inherits a mixture of the two opposing roles.

Ablation, transplant, and additive steering are different operations. The W/C identification has different downstream consequences depending on the intervention. We separate three operations the literature often blurs.

Ablation (zero out canceller outputs during the forward pass) raises the correct logit by $+0.13$ –

0.29 nats and shifts accuracy directionally $+2$ – 7 pp in 6/6 main cells (Figure 5; App. D.3). This is the cleanest actionable consequence of the W/C split.

Transplant (replace head outputs with their task-means) goes the *opposite* direction: writer-only transplant underperforms FV-mean transplant in accuracy in 6/6 cells (App. D.2). The reason: FV-mean transplant incidentally regularises cancellers’ per-prompt outputs, while writer-only leaves cancellers active.

Additive steering (inject a mean vector at the readout token) gives a larger absolute held-out logit shift for $v_{\mathcal{W}}$ than for v_{FV} in 6/6 main cells (App. D.1), but α is tuned on logit shift; under norm-matched injection with held-out α the partition effect on accuracy is not established (see [Limitations](#)). The lesson: population-level identification says which heads to intervene on, not that an aggregated steering vector is the right tool to do so; the per-prompt nature of canceller behaviour means writer-only loses to FV-mean transplant in 6/6 cells under any α -free criterion, with the gap predicted by the canceller fraction in Todd’s top- K (App. D.2).

Conclusion. The W/C split is structural, not prescriptive; it is invisible to magnitude-only ranking and inherited by any downstream FV-based aggre-

gation.

Limitations

Task scope. The primary analysis uses two synthetic ICL rule-following families (hierarchical, modular) with shared 4-shot surface form. Cross-template transfer extends to two vocabulary-ICL templates (antonym, country-capital). All tasks are short, closed-label, and in-context-learnable from ≤ 5 demonstrations; we do not test long-form generation, open-ended NL classification at scale, or instruction-following.

Model coverage. Six Pythia main cells (410M, 1B, 1.4B $\times$ hier, mod) carry the full statistical pipeline. Extension cells (Pythia-2.8B/6.9B/12B; Qwen2.5- $\{1.5, 7\}$ B; GPT-2-medium) use a lighter protocol with $n_{\text{eval}} = 200$. We do not test Meta’s Llama series or instruction-tuned variants (Qwen2.5 shares the Llama-style attention architecture but is a separate model family). One main cell (mod-1.4B) is the boundary case on sign-shuffle ($p_{\text{emp}} = 0.104$); two extension cells (6.9B mod, 12B hier) shift to PARTIAL (one CI grazes 0).

Steering claims are limited. We make no positive claim that v_W improves practical steering. The only α -free comparison (transplant, App. D.2) shows writer-only is at best tied with FV-mean on accuracy, with the gap predicted by Todd’s top- K canceller fraction. Held-out α on the 6 main cells (App. D.1) is on logit shift only; accuracy under held-out α with norm-matched injection and path-patching W/C across all 15 cells is left to future work.

Mechanistic case study has narrow generality. L11.H4 (Pythia-410M) is the dominant canceller in both rule cells, but its “distributed-bias content reader” characterisation is established only at this one head. We do not claim the case-study mechanism generalises to the full 27 canceller (cell, head) pairs; we do show population-wide that rank-1 copy-suppression holds for 0/27 pairs, V-cascade is ruled out in 4/6 cells (2/6 active, 2/6 vacuous), and per-source DLA is content-dominated for 20/27 (74%) cancellers.

Statistical scope. Cross-cell aggregation uses Holm-Bonferroni FWER at $\alpha = 0.05$ over the 6 Pythia main cells (5/6 reject). The verdict matrix (six independent tests, App. B.2) reports per-test BH-FDR within each test family but not family-

wise across the matrix as a whole, so the matrix is presented as a per-test pass-rate summary rather than a single composite test.

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A Methods and reproducibility

A.1 Statistical inventory

Every test the paper relies on. Bootstraps are paired-prompt percentile bootstraps with $B = 10,000$ unless noted; BH-FDR at $q = 0.10$ on raw p -values; display p -values floored at 1×10^{-6} .

A.2 Reproducibility

The full implementation and reproducibility pipeline are available at <https://github.com/henryhyw/function-vectors-two-populations>.

Models. Pythia checkpoints EleutherAI/pythia-410m-deduped, pythia-1b-deduped, pythia-1.4b-deduped; final training step, fp32, attn_implementation=eager.

Prompts. 4-shot template $f_0, f_1, Y \dots f_{0q}, f_{1q}$, with $f_0, f_1 \sim \text{Unif}\{0, \dots, 7\}$, $z \sim \text{Unif}\{0, 1, 2, 3\}$; format-prefix is the literal ", ". Per cell: 192 DLA prompts (seed 42), 200 PP prompts (seed 43), 500 eval prompts (seed 44). Logit aggregation: $\log \sum p$ over leading-space variants $\{ 'A', 'A' \}$ vs. $\{ 'B', 'B' \}$.

FV head set. Sign-preserving refined-DLA passes at $q = 0.10$ unioned with top- K by $|\Delta\ell|$, $K = \text{clip}(2n_{\text{FDR}}, 8, 50)$, intersected with PP gate $|\text{direct\%}| \geq 5\%$ to yield $\mathcal{F}$.

Path patching. Direct-path receiver-edge ablation of Wang et al. (2023); filter $|\Delta\ell^{\text{full}}| < 0.5$.

QK bucketing. Five deterministic bucket spans: BOS = $\{0\}$; format prefix (whitespace/punctuation between demos); demonstration input ((f_0, f_1) spans); demonstration label (Y tokens); query input ((f_{0q}, f_{1q})).

Edge-level null. 50 random head pairs (L_a, L_b) with $L_a < L_b$ from the full head population per cell.

Table 6: **Statistical inventory.**

Test	Null	Statistic	p -floor	Multcomp
Refined-DLA	20 label permutations	per-head signed mean	0.05	BH $q=0.10$
Path patching	$B=10,000$ paired bootstrap	ratio-of-means / two-sided	$1/B$	BH $q=0.10$
Group lesion	$B=10,000$ paired bootstrap	4-condition CI gate	$1/B$	—
Sign-shuffle null	$n_{\text{perm}}=10,000$ in-pool sign	$ W-C $ contrast	10^{-4}	BH $q=0.10$
Within-layer OV null	≥ 100 same-layer non-($W \cup C$) pairs	cosine $\leq$ obs	0.01	—
Edge-null ACDC	50 random layer-respecting pairs	MWU on $ \Delta\ell $	1×10^{-6}	BH $q=0.10$
Specificity ratio	$B=5,000$ canceller bootstrap	rule/random NLL ratio	—	—
Induction TOST	hypergeometric E , margin $\sqrt{\text{Var}_{\text{hg}}}$	two one-sided test	—	—
Cross-task verdict	sign+magnitude on dst eval (relaxed gate, §3.5)	4-condition variant	—	—
Split-half	$n_{\text{split}}=5$ random 50/50 splits	verdict-stable in $\geq n/2$ splits	—	—
Per-source DLA	per-token-source bucket attribution	sink vs. content DLA share	—	—
Vocab transfer	$B=2,000$ paired bootstrap, $n=100$	$\Delta\ell^{\text{correct}}$ shift / sign agreement	$1/B$	—
Cochran–Armitage	rule-specific rate by ordinal scale	trend z , two-sided	—	—

Statistics. Bootstrap $B = 10,000$; sign-shuffle $n_{\text{perm}} = 10,000$; refined-DLA 20-seed permutation null; specificity $B = 5,000$. Thresholds ($\pm 5\%$ direct effect, 0.10 nat floor, $5\times$ rule/random ratio) fixed pre-analysis.

Computational budget and software. The released reproduction package is a single-notebook pipeline with modular Python stages for prompt generation, refined DLA, path patching, group lesion, QK-source attribution, rule-outs, transfer, scale extension, case study, and aggregation. It uses `torch`, `transformer_lens`, `transformers`, `numpy`, `scipy`, and `matplotlib`. A full regeneration of all released results takes approximately 18 hours on one NVIDIA A100 GPU; discovery and path patching dominate the runtime, with Pythia-6.9B and Pythia-12B accounting for most of the extension-cell cost. The released `results/directory` and `extracted_numbers.json` provide the seeded outputs used for all paper-cited numbers, so rerunning the notebook is not required to inspect the reported results.

Artifacts, licenses, and intended use. We use and release scientific artifacts: deterministic prompt generators, analysis code, figure scripts, per-cell JSON results, auxiliary result payloads, and an aggregation file containing every paper-cited number. The released code is MIT-licensed. Pre-trained model artifacts retain their original terms: Pythia and Qwen2.5 under Apache 2.0, GPT-2 under MIT. The intended use of the released package is research reproduction and inspection of the mechanistic claims in this paper; it is not intended as a deployment or safety-control system.

Data, human subjects, and risk profile. No human subjects or annotators are used. The rule-learning prompts are synthetic strings generated from integer feature pairs and deterministic label rules; the vocabulary-transfer checks use fixed common-word pairs such as antonyms and country-capital pairs. We did not collect private, personal, or newly authored human text data. The main foreseeable risk is technical over-interpretation or misuse of head-level interventions as a general model-control method; accordingly, the paper limits its claims to short closed-label ICL settings and explicitly separates population identification from practical steering.

A.3 Threshold sensitivity

Headline 11/12 canonical is robust to $\pm 50\%$ perturbation of any single threshold ($\pm 5\%$ direct-effect gate, 0.10-nat magnitude floor, $5\times$ rule/random ratio); the single boundary case (mod-1.4B mean, canceller +0.09) flips to canonical at floors ≤ 0.09 .

B Discovery and validation

B.1 Refined DLA and path patching: per-cell tables

Full per-head refined-DLA and PP tables released as JSONs; cell sizes in Table 3. End-to-end sum-to- $\Delta\ell$ reconstruction passes (rel. tol. 5%) in 9/9 tests for 6/6 cells.

B.2 Group lesion: zero and mean strategies

Table 7 mirrors Table 3 under mean ablation: 5/6 canonical, mod-1.4B writer-dominant (canceller +0.09 just below the 0.10 threshold).

Table 7: **Group lesion (mean strategy)**. 11/12 canonical across both strategies; the only miss is mod-1.4B mean.

Task	Model	$\Delta\ell_W$	$\Delta\ell_C$	$\Delta\ell_{\text{both}}$	verdict
hier	410M	-0.34	+0.29	-0.17	canonical
hier	1B	-0.44	+0.28	-0.34	canonical
hier	1.4B	-0.37	+0.22	-0.24	canonical
mod	410M	-0.31	+0.13	-0.23	canonical
mod	1B	-0.40	+0.26	-0.25	canonical
mod	1.4B	-0.35	+0.09	-0.26	writer-dominant

FV-set sign-shuffle null per-cell. Table 8: observed $|W-C|$ contrast vs. in-pool sign-shuffle null with Gaussian back-stop and BH- q . 5/6 cells reach $q_{\text{emp}} \leq 0.014$; mod-1.4B is the boundary ($p_{\text{emp}} = 0.104$, $p_{\text{Gauss}} = 0.088$).

Per-cell verdict matrix: FWER. Body Table 4 reports the per-cell verdict matrix. Headline FWER: Holm-Bonferroni on sign-shuffle p -values $\{0.001, 0.001, 0.001, 0.014, 0.014, 0.104\}$ at family-wise $\alpha = 0.05$ rejects 5/6 cells; mod-1.4B fails the final threshold. BH-FDR $q = 0.10$ yields the same conclusion. Other matrix entries are conjunctive threshold tests at fixed gates and reported descriptively.

B.3 Asymmetric budgets and per-task breakdown

Per-head $|\text{direct}\%|$: $W = 20.3\%$ ($n = 54$), $C = 11.2\%$ ($n = 27$); per-head canceller shift is more uniform than writer (CV 0.19 vs. 0.56; canceller budget ~ 0.04 nats/head, task-independent). Modular elicits $\sim 1.7\times$ larger writer magnitudes than hierarchical ($|\Delta\bar{\ell}_W| = 0.62$ vs. 0.38); cancellers not correspondingly amplified (0.17 vs. 0.20), so joint attenuation is lower on modular (36% vs. 61%).

B.4 ACDC edge ablation: pair-by-pair

Real-vs-null per cell: Mann-Whitney BH-FDR pass 6/6 at $q \leq 1.4 \times 10^{-11}$, rank-biserial $r \in [0.60, 0.97]$. Fisher canonical-rate BH-FDR pass: 1/6 (mod-410M, $q = 0.0016$; others raw $p \in [0.10, 0.21]$). Magnitude dominance is uniform; canonical-edge enrichment is task- and size-specific.

B.5 Scale and cross-architecture spot checks

C Mechanism rule-outs

C.1 Specificity controls: per-cell tables

Table 10: per-cell rule-specific / indeterminate / generally-important breakdown for 27 cancellers and the top-10 induction overlap. Rule-specific rate

sharpens with scale; no above-chance induction overlap. TOST equivalence margin is $\sqrt{\text{Var}_{\text{hg}}} \approx 0.5\text{--}0.7$ heads, so the test has limited power for 1-2 head intersections; the induction-distinction is also supported by QK-source attribution (§4.2).

Highest-specificity cancellers. L11.H4 (115 $\times$ hier-410M, 151 $\times$ mod-410M), L13.H1 (109 $\times$ hier-410M), L11.H2 (82 $\times$ mod-1.4B).

C.2 FV-overlap controls

Intersection of $\mathcal{F}$ with Todd et al. (2024)’s mean-ablation top- K at $K \in \{5, 10, 15, 20\}$ in Table 11. Total Jaccard 0.06-0.25 at top-20. The W vs. C asymmetry is task-dependent: hierarchical concentrates C in Todd’s top-20 (9/14 vs. writer 1/27); modular reverses (16/27 writers vs. 1/13 cancellers).

C.3 QK source distribution: per-cell attention mass

Per-cell mean attention mass by group and source bucket in Table 12. In 6/6 cells, demo-label mass drops from W to C (mean -0.13) and format-prefix rises (mean +0.13); other buckets shift by ≤ 0.05 .

C.4 Per-source DLA decomposition: cancellers vs. sinks

Per-canceller DLA attributed to source buckets via $\sum_{\text{pos}} \alpha_{\text{pos}} W_O W_V r_{\text{pos}}$ projected onto $u(x)$. Per-cell summary in Table 13; full JSON released.

Named-head detail. L11.H4: $\sim 100\%$ demo-label sourced in both cells.

C.5 Comparison to GPT-2 copy-suppression

Head-level detail behind body Table 5. Three load-bearing differences distinguish L11.H4 (this work) from GPT-2 L10.H7 (McDougall et al., 2023). *QK signature*: L10.H7 attends to copies of the query token; L11.H4’s attention is invariant across rule tasks but its DLA flips sign across templates (§4.7). *OV map*: L10.H7 has a rank-1 promote/suppress structure; L11.H4’s $W_O W_V$

Table 8: **FV-set sign-shuffle null with Gaussian back-stop** ($n_{\text{perm}} = 10,000$). Empirical and Gaussian p -values; BH-corrected q -values across the 6-cell family at $q = 0.10$. Empirical p floored at $1/n_{\text{perm}} = 10^{-4}$.

Cell	observed	null p_{50}	null p_{95}	z	emp p	Gauss p	q_{emp}	q_{Gauss}
hier-410M	-0.435	-0.01	+0.24	-2.79	0.0024	5.2×10^{-3}	0.007	0.016
hier-1B	-0.607	-0.10	+0.24	-2.47	0.0047	1.3×10^{-2}	0.007	0.027
hier-1.4B	-0.275	-0.04	+0.13	-2.32	0.0035	2.0×10^{-2}	0.007	0.031
mod-410M	-0.361	-0.04	+0.17	-2.08	0.0116	3.7×10^{-2}	0.014	0.045
mod-1B	-0.683	-0.03	+0.35	-2.87	0.0010	4.1×10^{-3}	0.006	0.016
mod-1.4B	-0.223	+0.01	+0.22	-1.71	0.1040	8.8×10^{-2}	0.104	0.088

Table 9: **Verdict across architectures and scale.** Group-lesion W / C / joint logit shifts (mean strategy, $n_{\text{eval}} = 200$). $W < 0$, $C > 0$ in all 9 extension cells; combined with the 6 main cells: 13/15 canonical, 2/15 partial (signed direction holds, one CI grazes 0).

Family / Arch	Size	Task	W_{shift}	C_{shift}	joint	verdict
Pythia (NeoX)	2.8B	hier	-0.30	+0.28	+0.01	canonical
Pythia	2.8B	mod	-0.20	+0.19	+0.01	canonical
Pythia	6.9B	hier	-0.15	+0.13	+0.02	canonical
Pythia	6.9B	mod	-0.11	+0.05	-0.01	partial
Pythia	12B	hier	-0.10	+0.17	-0.01	partial
Pythia	12B	mod	-0.11	+0.18	+0.11	canonical
Qwen2.5 (Llama-style)	1.5B	mod	-0.44	+0.11	-0.03	canonical
Qwen2.5	7B	mod	-0.11	+0.12	+0.10	canonical
GPT-2-medium (Conv1D)	355M	mod	-0.33	+0.38	+0.03	canonical

Table 10: **Per-cell specificity and induction-overlap controls.** *rule-spec*: cancellers with rule/random NLL ratio ≥ 5 . *ind. ovlp.*: $|\mathcal{F} \cap \text{top-10}_{\text{ind}}|/|\mathcal{F}|$ (intersection of FV head set with top-10 induction heads).

Cell	$ \mathcal{C} $	rule-spec	indet	gen-imp	ind. ovlp.
hier-410M	6	4	0	2	1/22
hier-1B	4	3	1	0	3/23
hier-1.4B	4	4	0	0	0/19
mod-410M	3	1	1	1	0/12
mod-1B	6	3	2	1	1/16
mod-1.4B	4	4	0	0	0/19
total	27	19 (70%)	4	4	—

Table 11: **Full 6-cell FV-overlap.** W/C set from the PP-validated set used throughout the paper; FV top- K from Todd’s mean-ablation identification on the same task. The W- vs C-asymmetry column shows W-in-FV / C-in-FV at $K = 20$; (C-heavy) marks cells where cancellers concentrate in Todd’s top-20, (W-heavy) marks the opposite.

Cell	$ \mathcal{W} $	$ \mathcal{C} $	Jaccard	W in FV-20	C in FV-20	pattern
hier-410M	8	6	0.06	0	2	C-heavy
hier-1B	9	4	0.18	1	4	C-heavy
hier-1.4B	10	4	0.10	0	3	C-heavy
mod-410M	8	3	0.19	5	0	W-heavy
mod-1B	8	6	0.17	5	0	W-heavy
mod-1.4B	11	4	0.25	6	1	W-heavy
hier total	27	14	—	1 (4%)	9 (64%)	—
mod total	27	13	—	16 (59%)	1 (8%)	—

top-1 Frobenius share is 2.8% with no rank-1 plateau (App. C.8). *Negation*: L10.H7’s behaviour under negation is not reported; L11.H4 flips canceller→writer on antonym, the expected behaviour of a “suppress what was demonstrated” mechanism. We do not yet test whether L11.H4

exhibits L10.H7’s full copy-suppression signature on copy-task contexts; this is left as future work.

Table 12: **Per-cell mean attention mass at the final query token, by W/C group and source bucket.** Each row is one (group, cell). n is the number of heads in the group. demo-lab/format-prefix differences are the load-bearing buckets of Figure 2.

Cell	Grp	n	BOS	format	d-in	d-lab	q-in
hier-410M	W	8	0.35	0.04	0.14	0.46	0.02
	C	6	0.34	0.15	0.11	0.33	0.07
hier-1B	W	9	0.41	0.11	0.09	0.35	0.03
	C	4	0.44	0.29	0.06	0.18	0.03
hier-1.4B	W	10	0.28	0.13	0.05	0.52	0.02
	C	4	0.27	0.26	0.06	0.37	0.05
mod-410M	W	8	0.34	0.17	0.11	0.36	0.02
	C	3	0.38	0.26	0.06	0.24	0.05
mod-1B	W	8	0.44	0.12	0.07	0.34	0.03
	C	6	0.46	0.20	0.07	0.22	0.06
mod-1.4B	W	11	0.32	0.13	0.06	0.45	0.03
	C	4	0.25	0.25	0.07	0.36	0.06

Table 13: **Per-cell content vs. sink decomposition of cancellers.** A canceller is content-driven if $|\text{BOS} + \text{fmt}| < |\text{demo}_{\text{in}} + \text{demo}_{\text{lab}} + \text{query}_{\text{in}}|$ in DLA contribution, sink-driven otherwise. $\sum \Delta \ell$ sums total per-head DLA over each sub-population. Content cancellers carry the headline magnitude in 5/6 cells; the hier-1B exception splits 2/4 heads as sink-classified. Stable cross-task sink heads: L14.H3 (both 410M), L10.H2 (both 1B), L11.H2 (both 1.4B).

Cell	content $ \mathcal{C} $	sink $ \mathcal{C} $	content $\sum \Delta \ell$	sink $\sum \Delta \ell$
hier-410M	5	1 (L14.H3)	-0.38	-0.03
hier-1B	2	2 (L10.H2, L12.H5)	-0.15	-0.13
hier-1.4B	3	1 (L11.H2)	-0.25	-0.04
mod-410M	2	1 (L14.H3)	-0.26	-0.02
mod-1B	5	1 (L10.H2)	-0.21	-0.02
mod-1.4B	3	1 (L11.H2)	-0.16	+0.00

C.6 Cross-template transfer to vocabulary ICL

Two vocab-ICL tasks (antonym, 60 pairs; country-capital, 50 pairs; $k=4$, format `word:answer`). Zero-strategy ablation, $n_{\text{eval}} = 100$, $B = 2,000$. Sub-population transfer in Table 14.

L11.H4 solo on vocab tasks. The *address* transfers but the *role* is task-conditional: on antonym L11.H4 acts as a writer ($\Delta \ell = -0.069$, 95% CI $[-0.109, -0.028]$); on country-capital it has no significant effect ($\Delta \ell = -0.005$, CI $[-0.033, +0.025]$).

C.7 Multi-cell mechanism rule-outs (rank-1 / V-cascade / V-shuffle)

Three named-mechanism rule-outs extended to the dominant canceller per cell: rank-1 (top-1 OV Frobenius share, all cancellers); V-cascade (zero-ablate top-3 anti-aligned upstream writers, $n_{\text{boot}} = 2000$ BCa); V-shuffle (relative collapse $\geq 0.5 = \text{content-driven}$).

Rank-1 ruled out population-wide (0/27 can-

cellers exceed 7.5% top-1 OV vs. 50% threshold). V-shuffle content-driven in 6/6 (rel-collapse 0.86–1.13). V-cascade ruled out 4/6; hier-1B and mod-1B show a small residual cascade ($\Delta \text{DLA} \in [+0.02, +0.03]$) from a near-orthogonal upstream writer (cosine $[-0.04, -0.01]$).

L11.H4 detailed CIs. Table 16 gives the paired-prompt bootstrap CIs for L11.H4 V-shuffle and V-composition (the case-study head also reported in §4.7). V-shuffle CIs exclude 0 in both cells (content-driven); V-composition CIs include 0 (parallel, not cascaded).

C.8 OV singular spectrum of L11.H4

D W/C aggregation: clean comparisons

We report two methodologically clean comparisons of v_{FV} vs. v_{W} . The (in-sample α , per-vector tuning, top- K heuristic W/C on extension cells) variant from an earlier pilot is excluded as confounded; results below use the path-patching W/C partition and either held-out α or α -free transplant.

Table 14: **Cross-template W/C transfer: 3/4 canonical pairs.** Mean shift in correct-token logit; positive = ablation helps correct; negative = ablation hurts. Canonical pattern requires (i) writers negative, (ii) cancellers positive, (iii) joint attenuates writers.

src cell	target task	writer $\Delta\ell$	canc $\Delta\ell$	joint $\Delta\ell$	verdict
hier-410M	antonym	-0.48	-1.56	-2.00	writer-dominant (fail)
hier-410M	country-capital	-0.52	+0.14	-0.37	canonical
mod-410M	antonym	-0.33	+0.03	-0.24	canonical (C-CI just over 0)
mod-410M	country-capital	-0.59	+0.23	-0.36	canonical

Table 15: **Multi-cell mechanism rule-outs.** $|C|$: cancellers in the cell. $\sigma_1^{\max}/\|OV\|_F^2$: max top-1 OV Frobenius share across all cancellers in the cell. $n_{\text{rank-1}}$: number of cancellers with top-1 OV ≥ 0.5 . $\Delta\ell$ V-shuffle: V-shuffle relative collapse on the top canceller. V-cascade verdict on the top canceller from ablating the top-3 anti-aligned upstream writers ($\cos < 0$): RULED OUT means CIs include 0 for all 3; PARTIAL means ≥ 1 writer’s CI excludes 0; VACUOUS means no upstream writer at $L < L_{\text{canceller}}$.

Cell	top canc.	$ C $	$\sigma_1^{\max}$	$n_{\text{rank-1}}$	$\Delta\ell$ V-shuf rel	V-cascade
hier-410M	L11.H4	6	0.070	0/6	0.86	RULED OUT
hier-1B	L13.H1	4	0.075	0/4	1.00	PARTIAL (L7.H2 $\cos = -0.04$)
hier-1.4B	L12.H7	4	0.030	0/4	0.98	RULED OUT
mod-410M	L11.H4	3	0.070	0/3	1.13	VACUOUS (no upstream W)
mod-1B	L13.H1	6	0.075	0/6	1.05	PARTIAL (L7.H0 $\cos = -0.01$)
mod-1.4B	L10.H7	4	0.030	0/4	0.95	VACUOUS (no upstream W)
cells		27	≤ 0.075	0/27	≥ 0.86 in 6/6	4/6 ruled out, 2/6 partial

Table 16: **V-shuffle and V-composition diffs on L11.H4 DLA.** V-shuffle: random permutation of per-head V across source positions ($n_{\text{eval}} = 200$, 3 seeds, paired-prompt bootstrap). V-composition: zero-ablate L10.H9 (dominant upstream writer).

Intervention	Cell	baseline DLA	post DLA	diff	95% CI
V-shuffle	hier-410M	-0.216	-0.040	+0.176	[+0.080, +0.279]
	mod-410M	-0.215	-0.103	+0.113	[+0.015, +0.206]
V-composition	hier-410M	-0.216	-0.209	+0.007	[-0.001, +0.015]
	mod-410M	-0.215	-0.211	+0.005	[-0.002, +0.011]

Table 17: **OV singular spectrum.** L11.H4’s $W_O W_V \in \mathbb{R}^{1024 \times 1024}$, $\|OV\|_F = 7.64$. The top-1 left-singular mode anti-aligns with task-mean $u(x)$ at $\cos = -0.13$, but its Frobenius share is only 2.8% (top-5: 13%; top-10: 25%); the map is dense, with no rank-1 plateau (McDougall et al., 2023).

mode k	σ_k	$\cos(u_k, u)$	signed contrib. $\sigma_k \cos$
0	1.27	-0.13	-0.16
1	1.26	+0.03	+0.04
2	1.24	+0.01	+0.01
3	1.23	+0.07	+0.09
4	1.21	-0.00	-0.00

D.1 Held-out α on 6 main cells

192/500 calibration-evaluation split; per-cell α selected on calibration over the grid $\{0.25, 0.5, 1, 2, 4, 8, 16\}$ to maximise $\overline{\Delta\ell}$, evaluated on disjoint hold-out. All three vectors injected at layer $\lfloor n_{\text{layers}}/2 \rfloor$ on the readout token.

D.2 Transplant accuracy: writer-only vs FV-mean

Each head’s activation slot is *replaced* with its task-mean ($n_{\text{eval}} = 200$). The asymmetry $\Delta\text{acc}(\mathcal{W}) - \Delta\text{acc}(\mathcal{F})$ is ≤ 0 in 6/6 cells, the opposite direction from the additive-injection logit-shift result. Replacement neutralises cancellers’ per-prompt outputs; sign-aware drop does not.

Table 18: Held-out logit shift at calibrated α .

Task	Model	$\Delta\ell(v_{\text{FV}})$	$\Delta\ell(v_{\text{W}})$	$\Delta\ell(v_{\text{PCA}})$
hier	410M	-0.002	-0.003	-0.007
hier	1B	-0.002	-0.007	-0.003
hier	1.4B	-0.004	-0.022	+0.003
mod	410M	+0.095	+0.159	-0.002
mod	1B	-0.010	+0.012	-0.002
mod	1.4B	-0.005	+0.097	-0.032

Table 19: Transplant accuracy ($n_{\text{eval}} = 200$).

Cell	baseline	$\Delta\text{acc}(\mathcal{F})$	$\Delta\text{acc}(\mathcal{W})$	asymmetry
hier-410M	0.525	+0.025	-0.040	-0.065
hier-1B	0.595	+0.000	-0.100	-0.100
hier-1.4B	0.600	+0.005	-0.110	-0.115
mod-410M	0.480	+0.090	+0.085	-0.005
mod-1B	0.515	+0.065	+0.060	-0.005
mod-1.4B	0.520	+0.085	+0.065	-0.020

D.3 Per-cell ablation accuracy effects

Group-lesion accuracy effects on the 6 main cells. Sign of the shift matches the W/C direct-effect sign in 6/6 cells; $\Delta\text{acc}(\text{W})$ CI excludes 0 in 3/6, $\Delta\text{acc}(\text{C})$ CI excludes 0 in 1/6.

D.4 Cancellers are writer-orthogonal in OV space

Decomposition $\bar{\mathbf{a}}_{\mathcal{C}} = \alpha_{\parallel} \bar{\mathbf{a}}_{\mathcal{W}} + \bar{\mathbf{a}}_{\mathcal{C}}^{\perp}$; perpendicular fraction 0.84–1.00 (mean 0.96, 95% CI [0.91, 0.99]); $\cos$ mean -0.18, 95% CI [-0.36, -0.03].

E Robustness controls

E.1 Head-randomised control: the tautology rule-out

Tests whether the canonical 4-condition verdict is a tautology of sign-partitioning a high-magnitude population. Per cell: $n_{\text{eval}} = 100$, top- $K = 14$ FV by $|\text{DLA}|$ vs. 10 random seeds of $|\text{DLA}|$ -matched non-FV partitions, zero-strategy group lesions on both. Table 22 reports per-cell aggregates.

FV ablations are $5.5\times$ (W) and $5.2\times$ (C) larger than $|\text{DLA}|$ -matched random, attenuation gap $2.5\times$ (0.047 vs. 0.019 nats). The sign part is mechanically forced; the loose attenuation flag fires for 77% of random seeds, so magnitude (not sign) distinguishes FV from non-FV. Matching is rank-closest (non-FV $|\text{DLA}|$ averages 0.04 vs. FV 0.20 nats); the mod-1.4B FV/random gap < 1 is consistent with its weak canceller pool ($|\mathcal{C}| = 4$).

Table 20: Δacc with Wald 95% CI.

Cell	baseline	$\Delta\text{acc}(W)$	$\Delta\text{acc}(C)$	$\Delta\text{acc}(W \cup C)$
hier-410M	0.55	-0.05 [-0.11, +0.01]	+0.07 [+0.01, +0.13]	+0.03 [-0.03, +0.09]
hier-1B	0.58	-0.16 [-0.22, -0.10]	+0.02 [-0.04, +0.09]	-0.12 [-0.18, -0.06]
hier-1.4B	0.61	-0.07 [-0.13, -0.01]	+0.05 [-0.01, +0.11]	0.00 [-0.07, +0.06]
mod-410M	0.56	-0.01 [-0.07, +0.05]	+0.03 [-0.03, +0.10]	+0.08 [+0.02, +0.15]
mod-1B	0.55	-0.23 [-0.29, -0.17]	+0.04 [-0.03, +0.10]	-0.16 [-0.22, -0.10]
mod-1.4B	0.56	-0.01 [-0.07, +0.06]	+0.04 [-0.03, +0.10]	-0.06 [-0.12, +0.00]

Table 21: Writer/canceller OV geometry on the 6 main cells.

cell	$\cos(\mathcal{W}, \mathcal{C})$	perp. frac.	$\alpha_{\parallel}$
hier-410M	+0.05	1.00	+0.08
hier-1B	-0.32	0.95	-1.02
hier-1.4B	+0.03	1.00	+0.04
mod-410M	-0.54	0.84	-1.91
mod-1B	-0.24	0.97	-0.22
mod-1.4B	-0.09	1.00	-0.26

Table 22: **Head-randomised control across 6 Pythia cells.** For each cell we report the FV-set’s writer/canceller/joint ablation logit shifts and the mean over 10 random $|\text{DLA}|$ -matched non-FV partitions. Sign convention: writer = $\text{DLA} > 0$ (pushes correct $\uparrow$), canceller = $\text{DLA} < 0$ (pushes correct $\downarrow$); ablating writers shifts $-$, ablating cancellers shifts $+$. Magnitudes $|W|$, $|C|$, gap = $|W| + |C| - |\text{both}|$. Last column: fraction of random matched seeds whose joint ablation passes the loose binary attenuation criterion $|\text{both}| < |W| + |C| - 0.005$.

Cell	FV-set			Random matched (mean over 10 seeds)				FV/rand gap
	$ W $	$ C $	gap	$ W $	$ C $	gap	atten frac	
hier-410M	0.184	0.068	0.034	0.037	0.034	0.008	8/10	4.3×
hier-1B	0.161	0.206	0.101	0.039	0.019	0.019	6/10	5.3×
hier-1.4B	0.097	0.097	0.082	0.046	0.031	0.042	9/10	1.9×
mod-410M	0.166	0.349	0.013	0.019	0.007	0.005	6/10	2.6×
mod-1B	0.148	0.311	0.045	0.015	0.043	0.017	8/10	2.6×
mod-1.4B	0.102	0.172	0.010	0.060	0.031	0.023	9/10	0.4×
mean	0.143	0.200	0.047	0.036	0.027	0.019	46/60 (77%)	2.5×